Decision and Social Choice

PE3101P — Lecture 2: Probability

Tomi Francis

National University of Singapore

20th August, 2026

Today

1. Logistics
2. Recap
3. Probabilities
4. Probabilism
5. Conditional probability
6. Dutch books

The week 1 exercise sheet is up

- On my website: `https://tomifrancis.com/pe3101p.html`

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- Optional, and not for credit.

A new document

I'm giving you the greatest gift a student can receive:

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- Optional — but may be useful for the for-credit assignment and other future material

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- Optional — but may be useful for the for-credit assignment and other future material
- In pdf and html. **I recommend the html**.

First for-credit assignment

- Goes out on my website **over the weekend**.

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- Due **Friday of Week 4 — 4th September**, by midnight.

My office

- I'm moving office today. From now on I'm in **AS3-05-01**.

My office

- I'm moving office today. From now on I'm in **AS3-05-01**.
- **Office hours today are 16:30–18:00**, later than usual — I'm moving in right after this lecture.

The register

- Printed this time. **Find your name and sign.**

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- It's in alphabetical order.

The register

- Printed this time. **Find your name and sign.**
- It's in alphabetical order.
- Can't find yourself? Then I don't have records of you yet — add your name and student number, then sign.

Two senses of "ought"

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What will *in fact* be best.

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What your **evidence indicates** will be best, on the balance of probabilities.

- Run across the road without looking and survive: correct in the fact-relative sense, a mistake in the evidence-relative one.
- **Decision theory is about the evidence-relative sense.**

Acts, states, outcomes

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- **States of nature**: the columns — ways the world could be, which you're uncertain about.
- **Outcomes**: the cells — how things turn out.

Conditions on the states — what I gave you

(i) Together with the act chosen, they **resolve the uncertainty** and guarantee one particular outcome.

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(i) Together with the act chosen, they **resolve the uncertainty** and guarantee one particular outcome.

(ii) The act you choose **does not affect** whether the state occurs, or how probable it is.

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These two are about the **set** of states, not each state on its own.

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Like rolling a die: one number ends up face up.

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- But **objective chances** are different: they're out in the world.
- Frequencies, or what a theory like quantum mechanics puts out.
- **Credences are in the agent's head** — how likely she thinks something is, given her evidence.

Utility and expected utility

A fair coin. **Take the bet**: \$100 on heads, nothing on tails.
Decline: \$50 either way.

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- Where do the numbers come from? Later in the course. For now: **more is better**.
- The **expected utility** of an act: the utilities of its outcomes, weighted by the probabilities of the states.

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Better in **some** state, at least as good in the rest.

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Strong statewise dominance

Better in **all** states.

Bayesianism: two constraints

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Today: both of them.

The sample space

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All the possible outcomes — all the states of nature you're uncertain about in the situation at hand.

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A sample space is just a **set**: a collection of things.

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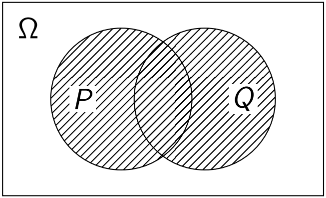
- The event with **no** outcomes in it is $\varnothing$: the **empty event**. It can never happen.

Combining events

$P \cup Q$ — something in **either** happens



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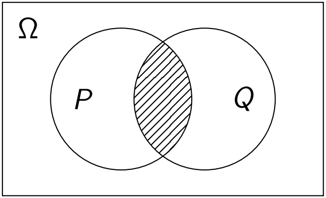
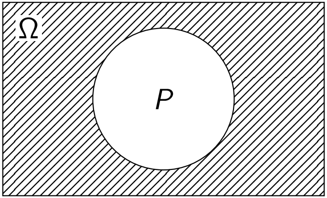


Combining events

P^c — everything in Ω **outside** P



We'll often write $\neg P$ for this.

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I won't penalise you for writing $\wedge$ instead of $\cap$.

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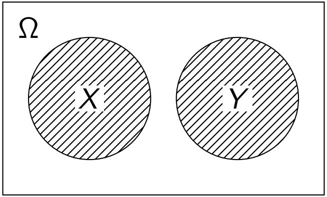
We'll assume any set of outcomes is an event.

Mutually exclusive

X and Y have **no outcome in common**: $X \cap Y = \varnothing$



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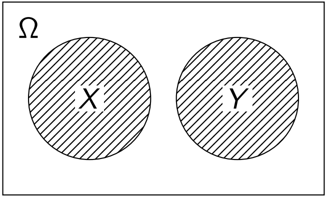
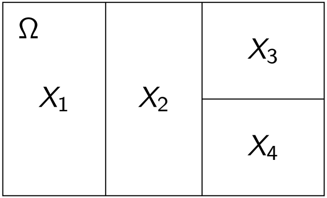


Also called **disjoint**.

Partitions

Jointly exhaustive: $X_1 \cup \cdots \cup X_n = \Omega$



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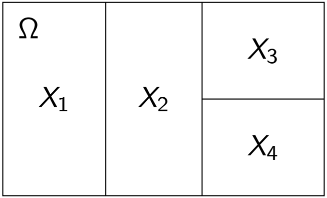

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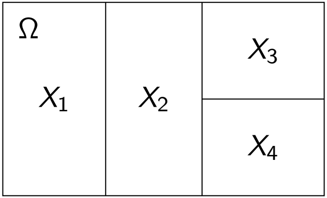

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Exactly one of them obtains — like the states of a decision table.

What's wrong with these?

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Homer thinks there's a 30% chance he'll drink one beer, a 50% chance he'll drink more than one, and a 70% chance he'll drink at least one.

The axioms — informally, as last week

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A die lands 1 *or* 2 with probability $1/6 + 1/6 = 1/3$.

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(iii) if $X \cap Y = \varnothing$, then $\text{Cr}(X \cup Y) = \text{Cr}(X) + \text{Cr}(Y)$

So what went wrong?

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- **Biqing.** Failing badly is a *way of* failing. So it can't be more probable than failing.
- **Homer.** One beer and more than one are disjoint, and together they are "at least one". So that should be $0.3 + 0.5 = 0.8$ — not 0.7.

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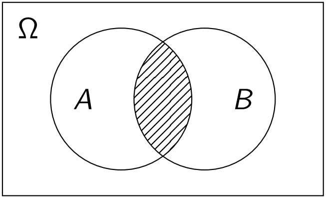
Note the proviso: conditioning on something you've ruled out is **undefined**, not zero.

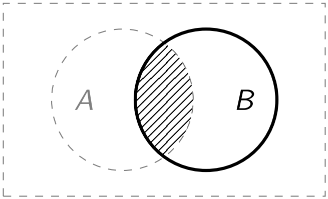
Why that formula?

Before: Ω is everything.



Why that formula?

After: B **is everything.**



The shaded region is $A \cap B$ both times. What changes is **what it's measured against**.

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Learning B tells you **nothing** about A.

Conditionalisation

The Principle of Conditionalisation

Suppose your credences are Cr, and you learn E **and nothing more**. Then your new credence in any P should be

$$\mathrm{Cr}_{\mathrm{new}}(P) = \mathrm{Cr}(P \mid E).$$

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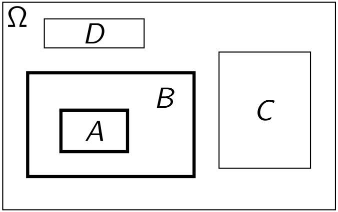
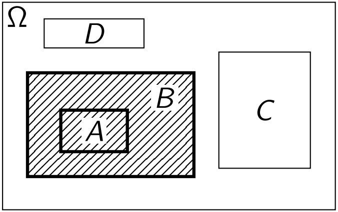
Your old **conditional** credence becomes your new **unconditional** one.

Conditionalisation, in pictures



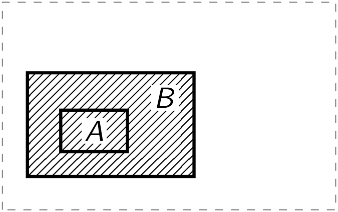
Areas are probabilities. We're going to condition on B.

Conditionalisation, in pictures



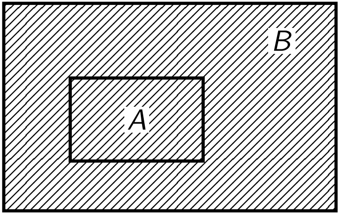
B is the **conditioning event**.

Conditionalisation, in pictures



Everything outside B is **discarded**. Nothing has moved yet.

Conditionalisation, in pictures



B **blown up** to fill the space Ω used to. Every ratio inside is unchanged.

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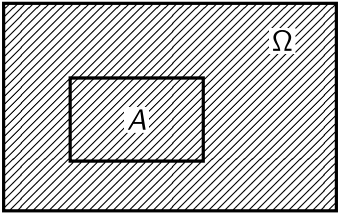
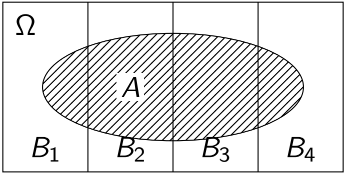


B is the new sample space.

Total probability



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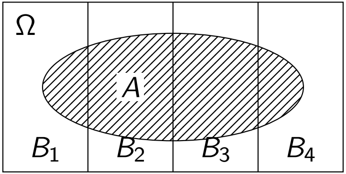

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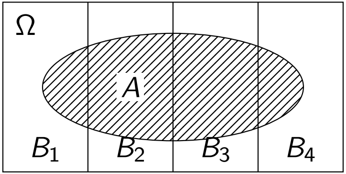

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Their probabilities **add up to** $Cr(A)$.

Bayes' theorem: what we have

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$$\mathrm{Cr}(A \mid B) = \frac{\mathrm{Cr}(B \mid A)\ \mathrm{Cr}(A)}{\mathrm{Cr}(B)} \qquad \text{provided } \mathrm{Cr}(A), \mathrm{Cr}(B) > 0.$$

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Two hypotheses H_1, H_2, and some evidence E.

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A Bayes factor of 1 moves you not at all.

Some practice

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- $\mathrm{Cr}(A) = 0.5$, $\mathrm{Cr}(B) = 0.4$, $\mathrm{Cr}(A \cap B) = 0.2$. Find $\mathrm{Cr}(A \mid B)$ and $\mathrm{Cr}(B \mid A)$. **Are A and B probabilistically independent?**

Base rates

Medical screening

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A DNA sample matches the defendant. An innocent person matches with probability 1 in a million.

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"So there's only a one in a million chance he's innocent."

Linking belief to action

Write a **\$100 bet on** A for a deal paying \$100 if A is true, nothing otherwise.

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We'll assume this today. It's a big assumption. Complaints held over.

Non-negativity, in one bet

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She loses \$40 **whatever happens**.

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And if $Cr(\Omega) = 1.2$? She buys the same bet for \$110 and collects \$100. **Same \$10 loss.**

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So we'll need **more than one bet**.

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P and Q are **mutually exclusive**. Our agent has credence **0.2** in P, **0.2** in Q, but **0.8** in $P \cup Q$ — where additivity requires **0.4**.

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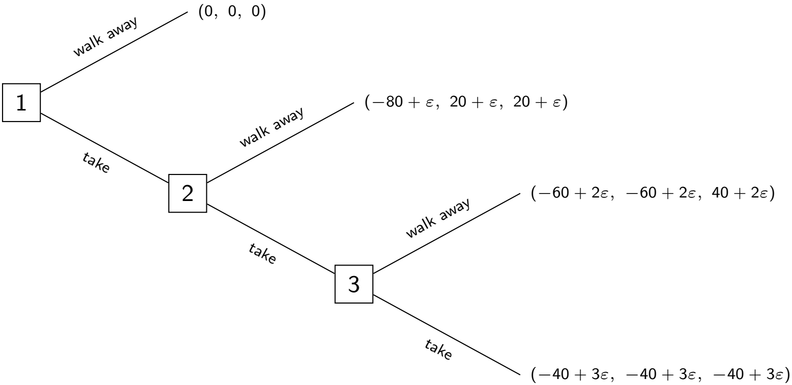
Bet 3 She **buys** a \$100 bet on $P \cup Q$ for \$$80 - \varepsilon$. Her price is \$80, so buying below it is a gain.

Each one, on its own, is a bet she **wants**.

But take all three. . .



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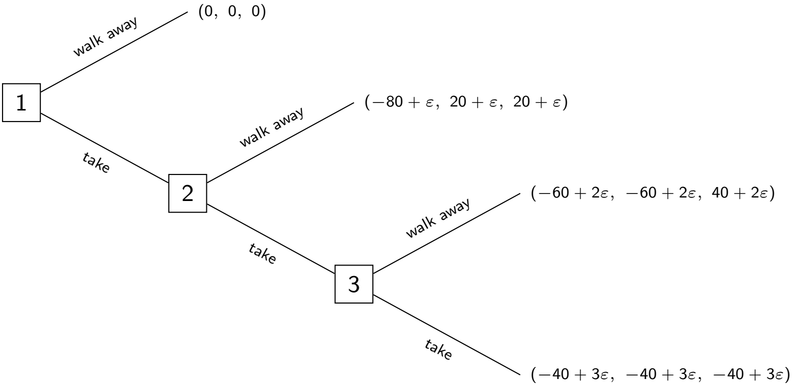

She is down about **$40**, however things turn out.

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So she walks away at node 1.

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Week 5, if there's time.

From the theorem to an argument

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- But there might equally be a **Frenchman** going around **rewarding** people with non-probabilistic credences.
- We wouldn't conclude you ought to have them. At most, that having them would be **practically useful**.
- And that's the problem: losing money is a **practical** misfortune. What was at issue is whether your beliefs are **epistemically** in order.

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1. You are rationally required to bet in line with the betting interpretation, given the credences you have.

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A better version

1. You are rationally required to bet in line with the betting interpretation, given the credences you have.
2. If you have non-probabilistic credences and bet in line with the betting interpretation, you are vulnerable to a Dutch book.
3. If you are vulnerable to a Dutch book, your credences are irrational.

So: if your credences are non-probabilistic, they are **irrational**.

The Frenchman

BDRC's objection: there might equally be a **Frenchman** going around **rewarding** people with non-probabilistic credences. So all the Dutch book shows is that probabilistic credences are **practically useful**.

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Making choices which **predictably** leave you worse off, for no reason, is irrational.

And it isn't about money

You're not rationally required to value money.

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The argument works for **any** quantity you do care about — so long as you're required to bet in line with the betting interpretation for it.

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- The weak premise is the **first** one: the betting interpretation of credence. You're obviously not required to maximise **expected money**.
- There's an answer: it's much more plausible when the stakes are **small**.
- And however small you make the stakes, you can **still** be made to lose for sure.

For next week

Newcomb's problem

BDRC ch. 9, and Adam Elga, "Newcomb University: A Play in One Act", *Analysis* 80.2 (2020), pp. 212–221.

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See you next week.